MODULE 31 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

Storage, Database and Observability Cost

Tier data by how it is used, provision performance separately from size, choose database capacity modes deliberately, and stop paying to keep logs nobody reads.

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Data costs grow quietly because nothing **deletes** it. Every store needs a rule for where data goes as it ages.

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S3 storage classes by access pattern

CLASS	FITS	WATCH FOR
Intelligent-Tiering	Unknown or changing access	Small monitoring fee per object
Standard-IA	Known infrequent, instant access	Retrieval fee, 30-day minimum
Glacier Instant Retrieval	Rarely read, milliseconds	90-day minimum
Glacier Flexible Retrieval	Archive, minutes to hours	90-day minimum
Glacier Deep Archive	Compliance archive, hours	180-day minimum

ON THE EXAM

Unknown access pattern: Intelligent-Tiering. Known and predictable: a lifecycle rule to the right class.

Lifecycle rules, done well

- **Transition** by age, prefix or tag; **expire** what no rule requires.
- **Clean up** old versions and incomplete multipart uploads.
- **Minimum durations**: early deletion still bills the minimum.
- **Tiny objects** cost more in IA and Glacier per object; aggregate first.
- **EFS lifecycle**: Infrequent Access and Archive tiers, with return on access.

> ON THE EXAM
>
> A versioned bucket growing without explanation: noncurrent versions with no expiry rule.

EBS volume choice

VOLUME	PERFORMANCE MODEL	FITS
gp3	IOPS and throughput set apart from size	Most workloads; default
gp2	IOPS grow with size, burst credits	Legacy; migrate to gp3
io2 Block Express	High provisioned IOPS, high durability	Critical, latency-sensitive databases
st1, sc1	Throughput-optimised HDD	Large sequential, cold data

ON THE EXAM

Over-sized gp2 volumes bought for IOPS: move to gp3 and provision the IOPS directly.

Database capacity choices

CHOICE	FITS
DynamoDB on-demand	Unknown or spiky traffic; pay per request
DynamoDB provisioned plus auto scaling	Steady, predictable traffic, high utilisation
Aurora Standard	Low or moderate I/O
Aurora I/O-Optimized	I/O over about a quarter of Aurora spend
Aurora Serverless v2	Variable load; can scale to zero
Reserved instances or Database Savings Plans	Steady database spend

ON THE EXAM

I/O charges dominating an Aurora bill: switch the cluster to I/O-Optimized.

Backup retention without waste

- **AWS Backup lifecycle** moves recovery points to cold storage, then deletes.
- **Data Lifecycle Manager** automates EBS snapshots and AMIs, with retention.
- **EBS Snapshots Archive** for long retention, rarely restored.
- **Retention by tier**: frequent short-term points, few long-term ones.
- **Cross-Region copies** only for what recovery needs.

> ON THE EXAM
>
> Seven-year snapshot retention, rarely restored: archive tier, not standard snapshots.

Cost-effective observability

LEVER	EFFECT
Filter at source	Do not ship debug logs nobody reads
Log group retention	Default is never expire; set it
Infrequent Access log class	Lower ingestion price for rarely queried logs
CloudWatch Logs intelligent tiering	Ageing logs move to cheaper storage tiers
Metric aggregation	Fewer dimensions, standard resolution, EMF
Trace sampling	Keep a representative sample, not every request

ON THE EXAM

Log groups with no retention setting keep everything forever and bill for it.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Access pattern unknown or changing	S3 Intelligent-Tiering
Kept years for compliance, hours to restore	Lifecycle to Deep Archive
Big gp2 volumes bought for IOPS	gp3 with provisioned IOPS
Aurora bill dominated by I/O	Aurora I/O-Optimized
Years of snapshots, rarely restored	Snapshot archive, lifecycle
Logs kept forever, rarely read	Retention, IA class, filtering

ON THE EXAM

Ask how data is used as it ages, then match the tier and the deletion rule.

> Meridian Freight's document store holds 400 TB of shipment files, mostly over 1 MB. Some files are read daily for weeks, others never again, and the pattern differs by customer and cannot be predicted.
>
> The team wants lower storage cost without retrieval surprises.

What should the solutions architect recommend?

A. A lifecycle rule moving everything to Standard-IA after 30 days

B. Move everything to Glacier Flexible Retrieval

C. S3 Intelligent-Tiering for the bucket

D. Keep everything in S3 Standard

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Standard-IA after 30 days

Files still read daily would incur retrieval fees.

B Glacier Flexible

Active files would take minutes to hours to read.

C Intelligent-Tiering

Tiers each object by its own access, with no retrieval fees.

D Standard

Pays the top rate for files nobody reads.

THE SIGNAL

Unpredictable access per object, large objects: Intelligent-Tiering.

> Meridian Freight must keep signed customs declarations for seven years. After 90 days they are almost never read, and regulators accept retrieval within 48 hours.
>
> The files are currently in S3 Standard.

What should the solutions architect configure?

A. S3 Intelligent-Tiering with default settings

B. A lifecycle rule to Standard-IA after 90 days

C. A lifecycle rule to Glacier Instant Retrieval after 90 days

D. A lifecycle rule to Glacier Deep Archive after 90 days, expiring after seven years

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Intelligent-Tiering

Pays monitoring fees on a pattern that is already known.

B Standard-IA

Far more expensive than an archive class for seven years.

C Glacier Instant

Pays for millisecond access nobody needs.

D Deep Archive, then expire

Cheapest class, retrieval within hours, deleted at the end.

THE SIGNAL

Known pattern, long retention, hours acceptable: lifecycle to Deep Archive, then expire.

> Meridian Freight's database servers use 2 TB gp2 volumes. The data is only 300 GB, but the volumes were enlarged years ago to get 6,000 IOPS.
>
> The team wants the same performance at lower cost.

What should the solutions architect recommend?

A. Move the data to new 400 GB gp3 volumes with 6,000 provisioned IOPS

B. Modify the volumes to io2 Block Express

C. Move the data to st1 volumes

D. Keep gp2 and buy a Savings Plan

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A gp3, right-sized, IOPS provisioned

Pays for 400 GB and the IOPS needed, not 2 TB.

B io2 Block Express

Premium performance the workload does not need.

C st1

HDD throughput volumes suit sequential data, not databases.

D Savings Plan

Savings Plans do not apply to EBS storage.

THE SIGNAL

Capacity bought to get IOPS: gp3, sized for data, IOPS provisioned separately.

> Cost Explorer shows that I/O charges make up 60% of the spend on Meridian Freight's busiest Aurora PostgreSQL cluster. Load is steady and the instances are correctly sized.
>
> The team wants to reduce the cluster's cost without changing the application.

What should the solutions architect recommend?

A Migrate to Aurora Serverless v2

B Switch the cluster's storage configuration to Aurora I/O-Optimized

C Move the database to self-managed PostgreSQL on EC2

D Add read replicas

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Serverless v2

For variable load; still charges per I/O on Standard.

B I/O-Optimized

No per-I/O charges; wins when I/O exceeds about a quarter of spend.

C Self-managed on EC2

Large operational cost and an application risk.

D Read replicas

More instances, and the I/O is still charged.

THE SIGNAL

I/O dominating the Aurora bill: I/O-Optimized.

> Meridian Freight keeps a daily EBS snapshot of each finance server for seven years, all in the standard snapshot tier, created and deleted by scripts that often fail.
>
> Restores of anything older than a month are rare, and a slower restore is acceptable for those.

What should the solutions architect implement?

A. Keep daily standard snapshots, and fix the scripts

B. Copy snapshots to S3 manually each month

C. Create AMIs instead of snapshots

D. A Data Lifecycle Manager policy keeping recent daily snapshots, archiving monthly ones to the snapshot archive tier, and deleting at seven years

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Fix scripts

Keeps seven years of daily points at full price.

B Manual copies

Manual, and snapshots cannot simply be copied to your own bucket.

C AMIs

Still snapshots underneath, at the same price.

D DLM with archive tier

Automated, tiered retention, archived long-term points, deletion.

THE SIGNAL

Long retention, rare restores: few long-term points, archived, managed by policy.

> Meridian Freight's CloudWatch Logs bill doubled in a year. Most log groups have no retention setting, services log at debug level in production, and most logs are only queried during audits.
>
> Logs must stay searchable for one year.

What should the solutions architect recommend?

A. Reduce production log level, set one-year retention on every log group, and use the Infrequent Access log class for audit-only logs

B. Export all logs to Glacier daily and delete them from CloudWatch

C. Turn off logging for most services

D. Move to high-resolution metrics instead of logs

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Level, retention, IA class

Less ingested, nothing kept past a year, cheaper ingestion for audit logs.

B Glacier export

No longer searchable, which is required.

C Turn off logging

Fails audit and operations needs.

D High-resolution metrics

Costs more, and metrics are not logs.

THE SIGNAL

Logs growing without limit: filter at source, set retention, choose the cheaper class.

What to carry out of this module

- **Unknown access:** Intelligent-Tiering. **Known:** lifecycle, then expire.
- **gp3**: size for data, provision IOPS separately.
- **Capacity modes:** on-demand vs provisioned; I/O-Optimized when I/O dominates.
- **Backups and snapshots:** tiered retention, archive, managed deletion.
- **Logs:** filter at source, set retention, use the cheaper class.

> ON THE EXAM
>
> Every store needs a rule for where data goes as it ages, and when it is deleted.